

Lecture 2 Live Problems

Quantification. State variables and change.

Points toward lab manual pages: **ST1, ST2, ST3, ST4** (section L1 S1)

1. Forrest Gump's hair. Tabulate your predictions for Forrest Gump's hair length H each month, given that he starts with a short 1 cm haircut. Then plot a time series from your table of predictions.

t (months)	0	1	2	3	4	5
H (cm)	1					



2. Write a change equation that reasonably models Forrest Gump's hair growth.

3. **Turkeys and coyotes, with numbers.** Tabulate your predictions for the populations of turkeys and coyotes over 5 months, if there are currently 1000 turkeys and 20 coyotes. Plot these in a time series and as a trajectory in state space.

t (months)	0	1	2	3	4	5
T	1000					
C	20					

